

# Mathematical Analysis of Modelling Head Collapse in MCTS Self-Play Training

Poker AI (ASI) — v6 architecture

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## 1. Notation and Computational Graph

Let  $\text{ctx}_0 \in \mathbb{R}^{1 \times L \times D}$  be the root perception output ( $L$  tokens,  $D = 384$ ). Define the heads:

$$\mathbf{M}(\cdot) : \mathbb{R}^{1 \times S \times D} \rightarrow \mathbb{R}^{1 \times A \times D} \quad (\text{modelling head, } A = 14 \text{ actions})$$

$$\mathbf{H}(\cdot) : \mathbb{R}^{1 \times S \times D} \rightarrow \mathbb{R}^A \quad (\text{action head: self-attn} \rightarrow \text{mean pool} \rightarrow \text{linear})$$

$$\mathbf{V}(\cdot) : \mathbb{R}^{1 \times S \times D} \rightarrow \mathbb{R} \quad (\text{value head: self-attn} \rightarrow \text{mean pool} \rightarrow \text{linear})$$

At chain depth  $d$ , the forward pass computes:

$$\mathbf{E}_d = \mathbf{M}(\text{ctx}_d) \in \mathbb{R}^{1 \times A \times D} \quad (\text{all action embeddings})$$

$$\mathbf{e}_d = \mathbf{E}_d[\mathbf{a}_d] \in \mathbb{R}^D \quad (\text{embedding of the taken action})$$

$$\text{ctx}_{d+1} = [\text{ctx}_d ; \mathbf{e}_d] \in \mathbb{R}^{1 \times (L+d+1) \times D}$$

$$\text{logits}_d = \mathbf{H}(\text{ctx}_{d+1}) \in \mathbb{R}^A$$

$$L_{\text{chain}}^d = \text{KL}(\text{softmax}(\text{logits}_d), \pi_d^{\text{MCTS}})$$

where  $\pi_d^{\text{MCTS}}$  is the visit-count distribution from search, and  $\mathbf{a}_d$  is the action actually taken at depth  $d$ .

With `stop_grad_old_embs=True`, when building  $\text{ctx}_{d+1}$  for depth  $d+1$ , the embedding  $\mathbf{e}_d$  is detached. Therefore, at depth  $d$  the gradient flows through  $\theta_M$  **only** via the current step's  $\mathbf{M}(\text{ctx}_d)[\mathbf{a}_d]$ .

## 2. Gradient Decomposition

### 2.1 Chain rule through the action head

The gradient of the chain KL loss at depth  $d$  w.r.t.  $\theta_M$ :

$$\frac{\partial L_{\text{chain}}^d}{\partial \theta_M} = \left( \frac{\partial L}{\partial \text{logits}_d} \right) \cdot \left( \frac{\partial \text{logits}_d}{\partial \text{ctx}_{d+1}} \right) \cdot \left( \frac{\partial \text{ctx}_{d+1}}{\partial \mathbf{e}_d} \right) \cdot \left( \frac{\partial \mathbf{e}_d}{\partial \theta_M} \right)$$

**Factor 1:**  $\partial L / \partial \text{logits}_d = \text{softmax}(\text{logits}_d) - \pi_d^{\text{MCTS}}$ . Standard KL gradient, a vector in  $\mathbb{R}^A$ .

**Factor 2:**  $\partial \text{logits}_d / \partial \text{ctx}_{d+1}$ . The ActionHead architecture processes  $\text{ctx}_{d+1}$  through 6 layers of causal Qwen3 self-attention, then applies **masked mean pool** and a linear projection  $\mathbf{W}_{\text{out}} \in \mathbb{R}^{A \times D}$ :

$$\mathbf{h} = (1 / (L+d+1)) \cdot \sum_{i=0}^{L+d} \mathbf{x}_i \quad (\text{mean pool over all positions})$$

$$\text{logits} = \mathbf{W}_{\text{out}} \cdot \mathbf{h}$$

**Factor 3:**  $\partial \text{ctx}_{d+1} / \partial \mathbf{e}_d$ . Since  $\text{ctx}_{d+1} = [\text{ctx}_d ; \mathbf{e}_d]$  and  $\text{ctx}_d$  is detached (`stop_grad_old_embs`), the gradient passes **only** through the last token position ( $L+d$ ). In the mean pool:

$$\frac{\partial \mathbf{h}}{\partial \mathbf{e}_d} = (1 / (L+d+1)) \cdot \frac{\partial \mathbf{x}_{L+d}}{\partial \mathbf{e}_d}$$

The causal self-attention means token at position  $L+d$  attends to all prior positions, but no subsequent position attends to it (it is the last). Therefore  $\mathbf{e}_d$  affects **exactly one** of the  $L+d+1$  terms in the mean pool.

### 2.2 Critical Scaling Factor: $O(1/(L+d+1))$

Combining factors 2 and 3, the effective gradient of logits w.r.t.  $e_d$  scales as:

$$\partial \text{logits} / \partial e_d = w_{\text{out}} \cdot (1/(L+d+1)) \cdot \partial x_{L+d} / \partial e_d$$

Concrete magnitude estimates:

| Scenario           | L (tokens) | d (depth) | L+d+1 | Scale factor |
|--------------------|------------|-----------|-------|--------------|
| Postflop, root     | 20         | 0         | 21    | 0.048        |
| Postflop, depth 3  | 20         | 3         | 24    | 0.042        |
| River, root        | 60         | 0         | 61    | 0.016        |
| River, depth 3     | 60         | 3         | 64    | 0.016        |
| Long hand, depth 5 | 80         | 5         | 86    | 0.012        |

Table 1. Mean-pool dilution factor for modelling head gradient.

Compare this with the root action loss, where  $H(\text{ctx}_0)$  uses the full  $L$ -token context and the gradient  $\partial L_{\text{root}} / \partial \theta_H$  has no  $1/L$  dilution (every token's gradient contributes with equal weight through self-attention). The modelling head receives a gradient that is **1–2 orders of magnitude weaker** than the action head's root gradient.

## 2.3 Reconstruction Loss Gradient

The reconstruction prediction (train.py lines 285–291) uses a **raw mean pool without self-attention**:

```
recon_pred = (1/(L+d+1)) * (sum_{i in ctx_d} x_i + e_d)
recon_target = mean_pool(perception(real_events_{d+1})).detach()
L_recon = MSE(recon_pred, recon_target)
```

The gradient w.r.t.  $e_d$ :

$$\partial L_{\text{recon}} / \partial e_d = (2/(L+d+1)) \cdot (\text{recon\_pred} - \text{recon\_target})$$

This suffers the **same  $1/(L+d+1)$  dilution**. But there is a deeper structural problem:

**The reconstruction target is action-agnostic.** The target  $\text{recon\_target} = \text{mean\_pool}(\text{perception}(\text{real\_events}_{d+1}))$  is determined by the actual game events at position  $d+1$  in the hand. This target is the **same** regardless of which action  $a_d$  was taken—it reflects what actually happened, not what would have happened under a different action. Therefore, the gradient  $\partial L_{\text{recon}} / \partial e_d$  pushes all embeddings (for all actions) toward the **same vector**:  $(L+d+1) \cdot \text{recon\_target} - \sum \text{ctx}_d$ .

This is a **collapse force**, not a diversification force. The reconstruction loss actively encourages  $e_a \approx e_b$  for all actions  $a, b$ .

## 3. Self-Reinforcing Target Degeneracy

### 3.1 MCTS with collapsed embeddings

During MCTS search (inference time, `@torch.no_grad`), the modelling head produces action embeddings that are stored in tree nodes (mcts.py line 614):

```
action_embedding=act_embs[0, a].detach()
```

Context at a leaf node is built by concatenating embeddings along the path (mcts.py line 534):

```
context = cat([root_ctx, stack(embeddings)], dim=1)
```

Suppose embeddings have collapsed:  $M(\text{ctx})[a] \approx e^*$  for all actions  $a$ . Then for any two paths of the same length from root to different leaves:

$$\text{ctx}(\text{leaf}_1) = [\text{root\_ctx} ; e^* ; e^* ; \dots] \approx \text{ctx}(\text{leaf}_2) = [\text{root\_ctx} ; e^* ; e^* ; \dots]$$

$\Rightarrow V(\text{ctx}(\text{leaf}_1)) \approx V(\text{ctx}(\text{leaf}_2))$ , so  $Q$ -values of all subtrees become approximately equal.

### 3.2 PUCT under uniform Q

The PUCT selection formula (mcts.py line 498):

$$\text{UCB}(a) = Q_a + c_{\text{puct}} \cdot P_a \cdot \sqrt{N_{\text{parent}}} / (1 + N_a)$$

With  $Q_a \approx Q_b$  for all  $a, b$ , the selection is determined **entirely** by the prior  $P_a$  and the visit count  $N_a$ . The resulting visit distribution  $\pi^{\text{MCTS}}$  converges to a smoothed version of the action head prior  $P$  (modulated by Dirichlet noise).

### 3.3 The positive feedback loop

This creates a closed-loop degeneracy:

- **Collapsed M**  $\rightarrow$  equal Q-values in tree  $\rightarrow \pi^{\text{MCTS}} \approx$  smoothed prior  $P$
- Chain KL target  $= \pi^{\text{MCTS}} \approx P \rightarrow$  gradient pushes action head to reproduce its own prior  $\rightarrow$  policy sharpens
- Sharper prior  $\rightarrow$  fewer actions explored in next cycle  $\rightarrow \pi^{\text{MCTS}}$  even more concentrated
- Gradient  $\partial L_{\text{chain}} / \partial \theta_M$  carries no action-differentiating information (target is the same regardless of which embedding was used)  $\rightarrow M$  stays collapsed

**Formally:** in the neighborhood of collapse  $e_a = e^* + \delta_a$ ,  $\|\delta_a\| \ll \|e^*\|$ :

$$\partial L_{\text{chain}} / \partial \delta_a = (1/(L+d+1)) \cdot J_{\text{attn}} \cdot W_{\text{out}}^T \cdot (\text{softmax}(\text{logits}) - \pi^{\text{MCTS}})$$

Since  $\pi^{\text{MCTS}}$  does not depend on  $\delta_a$  (targets are fixed from the search), and  $\text{softmax}(\text{logits})$  depends on  $\delta_a$  only through the  $O(1/(L+d+1))$  factor, the gradient has magnitude:

$$\|\partial L_{\text{chain}} / \partial \delta_a\| = O(1/(L+d+1)) \cdot \|\text{uninformative residual}\|$$

This gradient does not carry information about **which direction**  $\delta_a$  should deviate from  $e^*$ , because the target  $\pi^{\text{MCTS}}$  is the same for all  $\delta_a$ .

## 4. Stability of the Collapsed Fixed Point

We show that the collapsed state  $e_a = e^*$  for all  $a$  is a **stable fixed point** of the combined training dynamics.

**At the collapsed state, the total gradient on  $\theta_M$  is:**

$$\begin{aligned} \nabla_{\theta_M} L_{\text{total}} = \sum_d \gamma^d \cdot [ \\ & (\text{chain\_w}) \cdot O(1/(L+d+1)) \cdot g_{\text{chain}}^d \\ & + (\text{recon\_w}) \cdot O(1/(L+d+1)) \cdot g_{\text{recon}}^d \\ & + (\text{chain\_value\_w}) \cdot O(1/(L+d+1)) \cdot g_{\text{value}}^d \\ & ] \end{aligned}$$

where:

- $g_{\text{chain}}^d$  is uninformative (same target for all actions, as shown in Section 3.3)
- $g_{\text{recon}}^d$  is a **collapse force** (pushes all actions to the same vector, as shown in Section 2.3)
- $g_{\text{value}}^d$  is uninformative (V outputs a scalar, and the value target does not depend on which action embedding was appended)

All three gradient components either push toward collapse or carry no action-differentiating information. There is **no term** in the loss that creates a restoring force pushing embeddings apart.

Meanwhile, the much stronger root losses ( $\partial L_{\text{root}} / \partial \theta_{\text{perception}}$ ) update perception freely, which changes M's input distribution, adding noise to M's parameter updates without providing directional signal. This noise accelerates convergence to the degenerate fixed point.

**Conclusion:** The collapsed state is not merely stable—it is the **only attractor** in the current architecture. Every gradient component either pushes toward collapse or is neutral. No mechanism exists to separate embeddings once they approach each other.

## 5. Summary of Root Causes

| # | Cause                         | Mechanism                                                                                                                                      | Severity   |
|---|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1 | Mean-pool dilution            | Gradient to modelling head scales as $O(1/(L+d+1))$ because appended embedding is 1 of $L+d+1$ tokens in the mean pool. Factor: 0.01-0.05.     | Critical   |
| 2 | Action-agnostic recon target  | Reconstruction target = perception(real next events) is the same for all actions. Pushes ALL action embeddings toward the same vector.         | Critical   |
| 3 | Self-reinforcing MCTS targets | Collapsed embeddings => equal Q in tree => visit distribution reflects prior, not action quality => no action-differentiating gradient signal. | Critical   |
| 4 | No restoring force            | No term in the loss creates repulsion between embeddings. Collapsed state is the only attractor.                                               | Structural |

Table 2. Summary of root causes.

## 6. Implications for Fix Strategy

The analysis shows that the problem is **structural**, not a matter of hyperparameter tuning. Any fix must address at least two of the three critical causes simultaneously:

**Requirement 1:  $O(1)$  gradient magnitude.** The modelling head must receive a gradient that does not dilute with sequence length. This rules out architectures where the embedding is consumed solely through concatenation + mean pool.

**Requirement 2: Action-conditional targets.** The training signal must differ depending on which action's embedding is being evaluated. The reconstruction target must be action-specific: "what would the state look like if action a was taken" (counterfactual), not "what did the state look like" (factual).

**Requirement 3: Decouple from MCTS visit targets.** At least part of the modelling head's training signal must come from a source that is not derived from the MCTS search using the same (potentially collapsed) modelling head. The GTO phases already provide such a signal (action EVs from the solver), and the modelling phase (phase 4) uses a different loss. The question is how to maintain this during MCTS self-play.

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*Note: Diversity regularization (e.g., cosine-similarity penalty) addresses Requirement 3 (provides a collapse-independent signal) but not Requirements 1–2. It is a palliative that fights the symptom while the three critical forces still push toward collapse. Freezing perception stabilizes the recon target but kills the opponent GRU updates and still leaves Causes 1–3 intact.*